

An affirmative literature answer to the operator E -norm question in arXiv:1903.06086

Literature-status note for `fa_banach_001`

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Status. `literature_already_answered`.

Original question

Let G be an arbitrary positive densely defined operator on a separable complex Hilbert space $\mathcal{H}$, and let A be $\sqrt{G}$ -bounded on $\mathcal{D}(\sqrt{G})$. Shirokov's arXiv:1903.06086 asks, immediately after equation (13) on original PDF page 5, whether the mixed-state operator E -norm always has the simplified expression

$$\|A\|_E^G = \sup\{\|A\varphi\| : \varphi \in \mathcal{D}(\sqrt{G}), \|\varphi\| \leq 1, \|\sqrt{G}\varphi\|^2 \leq E\},$$

without assuming that G is discrete [1].

Exact later answer

The revised version of Shirokov's *Operator E -norms and their use* answers the question affirmatively [2]. Section 5 first defines $A\rho A^*$ for every finite-energy positive trace-class ρ and every $\sqrt{G}$ -bounded A . Proposition 6 on original PDF page 20 then states:

- the pure-vector and mixed-state definitions of $\|A\|_E^G$ coincide for every $\sqrt{G}$ -bounded A ; and
- both suprema may be taken over subnormalized competitors, specifically $\|\varphi\| \leq 1$ and $\text{Tr } \rho \leq 1$, with energy at most E .

This is exactly equation (13) of the question, with no discreteness assumption on G . The proof reduces arbitrary finite-energy states to finite-rank positive operators and uses the concavity of $E \mapsto \|A\|_E^{G^2}$; the tail then passes by monotone convergence of $\text{Tr}(A\rho_n A^*)$.

For an independent later cross-check, van Luijk's *Energy-Limited Quantum Dynamics* gives the subnormalized pure-vector formula as (2.33) and its equality with the mixed-state supremum as (2.34), in Lemma 2.19(2),(4) on arXiv PDF pages 15–16 [3].

Chronology and scope

The question appeared in arXiv:1903.06086v1 on 14 March 2019. The answering arXiv:1806.05668 was revised on 1 October 2020, after that question, and the published 2020 article contains Proposition 6. Thus the identifier of the answering work is numerically earlier, but the decisive revision is later. No new mathematical result is claimed by this packet.

References

- [1] M. E. Shirokov, *On extension of quantum channels and operations to the space of relatively bounded operators*, Lobachevskii J. Math. **41** (2020), 714–727; arXiv:1903.06086, <https://arxiv.org/abs/1903.06086>.
- [2] M. E. Shirokov, *Operator E -norms and their use*, Sbornik: Mathematics **211** (2020), 1323–1353; arXiv:1806.05668v6, <https://arxiv.org/abs/1806.05668>.
- [3] L. van Luijk, *Energy-Limited Quantum Dynamics*, Commun. Math. Phys. **406** (2025), article 120; arXiv:2405.10259, <https://arxiv.org/abs/2405.10259>.